What to Measure

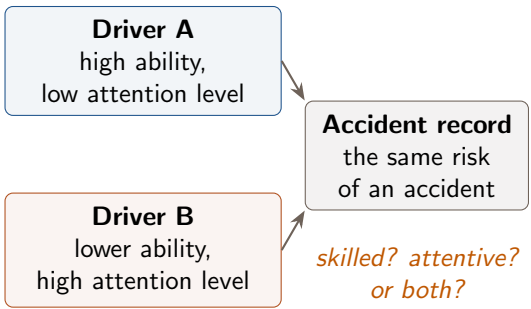
Optimal Monitoring under Adverse Selection and Moral Hazard

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One record, two reasons to be safe



- The insurer observes the **accident record**.
- Safety can come from ability (a private characteristic) or attention level (a hidden choice).
- The insurer uses contracts to **screen ability** and **encourage attention**.
- The same record can come from different combinations of ability and attention.

The questions

The problem. The accident record mixes ability and attention level.

the insurer can add one signal

1. What should a new telematics signal measure: ability, attention level, or a mix? Which signal direction is more valuable?

the signal changes the optimal contract

2. How does better monitoring change the allocation? Does it raise attention levels across the board, or reallocate attention across drivers?

allocation changes determine welfare

3. Who gains from better information, and when does more monitoring raise total surplus?

The plan

1. **The record-only benchmark**
   How a bonus encourages attention when ability is private.

2. **Find the optimal weak signal**
   Choose the signal direction and payments together. Then examine attention and welfare.

3. **Why the weak-signal calculation works**
   Four steps connect the benchmark to a general direction criterion. The detailed argument follows the takeaways.

The starting point

A weak signal changes the existing incentive problem only a little. Its value depends on which incentive requirements are costly at the benchmark.

Where this sits

- **Adverse selection + moral hazard:** Laffont–Tirole (1986, 1993), Picard (1987), Guesnerie–Picard–Rey (1989), Myerson (1982); Castro-Pires–Chade–Swinkels (2024), Gottlieb–Moreira (2022). *They take the information structure as given.*
- **Value and design of information in agency:** Holmström (1979), Kim (1995), Holmström–Milgrom (1991), Lewis–Sappington (1997), Georgiadis–Szentes (2020). *There a signal is about the action; here one signal serves two frictions.*
- **Optimal experimental design:** determinant-optimal designs and Fisher information (Kiefer 1959; Pukelsheim 1993). *The statistical criterion the economic one is compared with.*
- **First-order approach / local IC:** Mirrlees (1999), Rogerson (1985), Jewitt (1988), Rochet–Choné (1998). *Does the relaxation stay valid as information moves?*
- **Telematics and usage-based insurance:** Reimers–Shiller (2019), Soleymanian–Weinberg–Zhu (2019), Jin–Vasserman (2021). *Monitoring changes behavior and selects safer drivers; which data to collect is a design choice.*

Lessons from the model

- **Incentive needs shape what to measure.** Targeting costly incentive constraints can move the optimal direction away from the statistically optimal one.
- **Signal direction helps explain how contracts change.** With weak monitoring, rewards tied to the attention signal strengthen incentives for less able drivers and weaken them for some abler drivers.
- **Private gains need not translate into social gains.** The insurer gains and drivers lose in this example. Total surplus can rise or fall, depending on changes in accident losses, effort costs, and payment risk.

Benchmark: insurance without telematics

The driver

- Private ability $\theta \in [0,1]$. Higher θ means greater driving ability.
- Hidden attention level $a \in [0,1]$, at cost $c(\theta, a)$, increasing and convex in a, with $c(\theta, 0) = 0$.
- Risk-averse utility $u(t) - c(\theta, a)$ from a payment t. Outside option: 0.

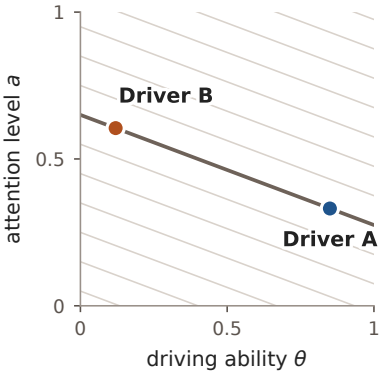
The insurer

- Observes only the **accident record** $y \in \{0,1\}$, where $y = 1$ means an accident.
- Bears a loss ℓ per accident and minimizes expected payments plus accident losses.
- Can offer transfers $t \geq 0$. Ability and attention are not contractible.

The accident record mixes ability and attention

Both greater ability and greater attention reduce accident risk:

$$p(\theta, a) = \Pr(y = 1 \mid \theta, a) = \Lambda(\gamma_0 - \gamma_\theta \theta - \gamma_a a), \qquad \Lambda(z) = \frac{1}{1 + e^{-z}}.$$



- The record depends on (θ, a) only through $\gamma_\theta \theta + \gamma_a a$.
- Driver A's greater ability offsets his lower attention. A and B therefore have the **same distribution of** y.
- A reward for avoiding an accident rewards both sources of safety. The insurer cannot tie it to attention alone.

The illustration uses $\gamma_\theta = 1.5$ and $\gamma_a = 4$. Gray lines join pairs with the same accident risk.

The menu and all incentive constraints

Write each transfer in **utility units**: $v_y = u(t_y)$. Its money cost is $h(v_y) = u^{-1}(v_y)$.

A report $\hat{\theta}$ selects utility payments $(v_0(\hat{\theta}), v_1(\hat{\theta}))$ and *deterministic* recommended attention $a(\hat{\theta})$. The driver chooses **both the report and actual attention**.

$$\min_{a, v_0, v_1} \mathbb{E}_\theta[(1 - p(\theta, a(\theta)))h(v_0(\theta)) + p(\theta, a(\theta))h(v_1(\theta)) + \ell p(\theta, a(\theta))] .$$

Full incentive compatibility requires

$$U(\theta) \geq (1 - p(\theta, \hat{a}))v_0(\hat{\theta}) + p(\theta, \hat{a})v_1(\hat{\theta}) - c(\theta, \hat{a}) \quad \text{for every } (\hat{\theta}, \hat{a}).$$

- **Truth-telling:** change the report, follow that report's recommendation.
- **Obedience (on-path):** report truthfully, change attention.
- **Joint deviations (obedience off-path):** take another contract and choose attention for one's true ability.

Payments and the driver's attention

outcome	utility payment	money transfer
accident ($y = 1$)	v_1	$h(v_1)$
no accident ($y = 0$)	v_0	$h(v_0)$

Facing (v_0, v_1), a driver of ability θ chooses $a \in [0, 1]$ to maximize

$$(1 - p(\theta, a))v_0 + p(\theta, a)v_1 - c(\theta, a).$$

- **The payment difference changes attention.** At an interior choice,

$$-p_a(\theta, a)(v_0 - v_1) = c_a(\theta, a).$$

- **The payment level changes utility.** Raising both utility payments equally leaves the attention choice unchanged.

What information could help?

Additional monitoring may improve the contract through two channels.

- **Information about attention**

  Better monitoring of hidden effort may allow the insurer to provide incentives at a lower cost.

- **Information about ability**

  Better identification of types may allow the insurer to offer different contracts and tailor incentives to individual drivers.

An additional signal may measure either dimension or a combination of both.

What should the insurer measure?

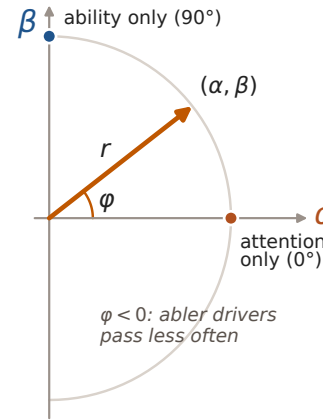
Signal design: precision and direction

The signal is **pass or fail**. Conditional on (θ, a), it is independent of the accident record:

$$q(\theta, a) = \Pr(\text{pass} \mid \theta, a) = \Lambda\big(\alpha(a - a_c) + \beta(\theta - \theta_c)\big).$$

Write $(\alpha, \beta) = r\omega(\varphi)$, where $\omega(\varphi) = (\cos\varphi, \sin\varphi)$.

- **Precision** r**:** how strongly the pass rate responds to the measured combination.
- **Direction** φ**:** the mix of attention and ability. 0° measures attention only, 90° ability only.
- A negative ability weight makes a more able driver less likely to pass at the same attention level.



Choosing a signal direction

The insurer chooses the direction of an additional signal with precision r.

The economic problem

1. Fix precision r and a candidate direction φ.
2. Reoptimize contracts and attention subject to full incentive compatibility.
3. Evaluate the insurer's value under the new signal.

Let $V(r, \omega(\varphi))$ denote the insurer's optimal expected payoff (negative cost), and let V_0 denote the benchmark without additional monitoring signal.

$$V(r, \omega(\varphi)) - V_0$$

measures the value of additional monitoring.

The local value of information

The first-order value of an uninformative signal vanishes (Radner and Stiglitz, 1984; Chade and Schlee, 2002).

If V is twice differentiable,

$$V(r,\varphi) - V_0 = r^2 \omega(\varphi)^\top M \omega(\varphi) + o(r^2).$$

The local information-design problem

$$\lim_{r \to 0^+} \max_{\varphi} \frac{V(r,\varphi) - V_0}{r^2}.$$

The matrix M determines which direction has the greatest local economic value.

Incentives and existing information

Local conditions implied by incentive compatibility

$$\underbrace{\begin{pmatrix}\partial_a \mathbb{E}[v(Y)] \\ \partial_\theta \mathbb{E}[v(Y)]\end{pmatrix}}_{\text{Marginal payment incentives}} = \underbrace{\mathbb{E}[v(Y)\lambda^Y]}_{\text{Score moments}} = \underbrace{\begin{pmatrix} c_a \\ U'(\theta) + c_\theta \end{pmatrix}}_{\Delta(\theta)}.$$

Here $\lambda^Y = \nabla_{(a,\theta)} \log f^Y(Y \mid \theta, a)$ is the score of the existing record.

Existing information

$$I^Y(\theta) = \mathbb{E}[\lambda^Y (\lambda^Y)^\top].$$

Information-adjusted incentives

$$\eta(\theta) = I^Y(\theta)^{-1} \Delta(\theta).$$

The same incentive requirements can be more or less costly to satisfy, depending on what the existing record reveals.

Characterizing the optimal direction

The local value matrix

In the regular characterization, write

$$M = \mathbb{E}_\theta[\eta(\theta)\eta(\theta)^\top], \qquad \eta(\theta) = I^Y(\theta)^{-1}\Delta(\theta).$$

For a signal direction $\omega = (\cos\varphi, \sin\varphi)^\top$,

$$\omega^\top M \omega = \mathbb{E}_\theta[(\omega^\top \eta(\theta))^2].$$

What does this mean?

- Each type has its own information-adjusted incentive direction $\eta(\theta)$.
- A signal is more valuable when its direction aligns with these incentive requirements.
- The insurer chooses one direction to balance these values across different types.

$$\omega^* \in \arg\max_{\|\omega\|=1} \omega^\top M \omega.$$

The optimal direction is a leading eigenvector of M.

Numerical illustration: a pooled benchmark

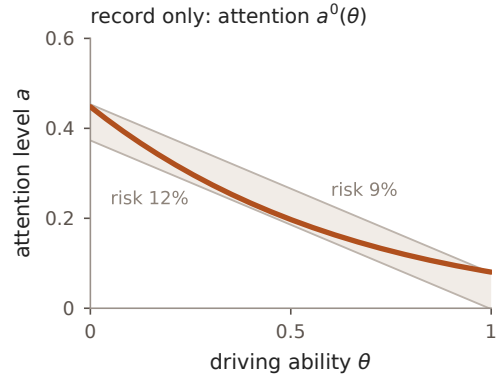
Specification

$$\theta \sim U[0,1], \qquad a \in [0,1],$$
$$c(\theta, a) = \tfrac{1}{2}(0.75 + \theta)^2 a^2,$$
$$p(\theta, a) = \Lambda(-0.5 - 1.5\theta - 4a),$$
$$u(t) = \frac{1 - e^{-0.5t}}{0.5}, \qquad \ell = 24.$$

Attention costs more for abler drivers.

Record-only benchmark

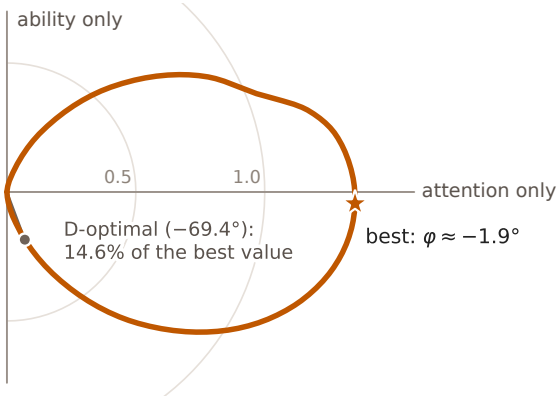
- **One contract:** $(v_0^0, v_1^0) = (0.76, 0)$, a \$0.95 reward after no accident.
- **Different behavior:** $a^0(\theta)$ falls from 0.45 to 0.08 as ability rises, so accident risk stays at about 9% to 12%.



The best weak direction

Local value of a weak signal in direction φ:

$$g(\varphi) = \lim_{r \to 0^+} \frac{V(r, \varphi) - V_0}{r^2}.$$



- **Best direction:** $\varphi^{\text{loc}} \approx -1.9°$, almost pure attention; $g \approx 1.351$.
- **Attention only** earns 99.9% of the best.
- **Economic vs. statistical:** the D-optimal direction, orthogonal to the record's index, earns only 14.6% of the best.

Why does the new signal help?

A more responsive rewarded outcome

	Record only	Record and signal
Rewarded outcome	No accident	No accident and pass
Probability	$1 - p$	$(1 - p)q$
Sensitivity to attention	$\gamma_a p$	$\gamma_a p + \alpha(1 - q)$

Sensitivity: the derivative of the log probability with respect to attention.

Two channels

1. **Attention incentives:** with $\alpha > 0$ the rewarded outcome responds more to attention, so each unit of expected utility paid buys a stronger marginal incentive.

2. **Screening and joint deviations:** a small negative ability weight makes passing less likely for abler drivers who claim a lower-ability contract without raising their attention.

Hence the best weak direction is close to, but not exactly, pure attention.

Beyond weak monitoring

Does the weak-signal direction survive stronger signals? For each precision r, we reoptimize the direction, the contracts, and the attention levels:

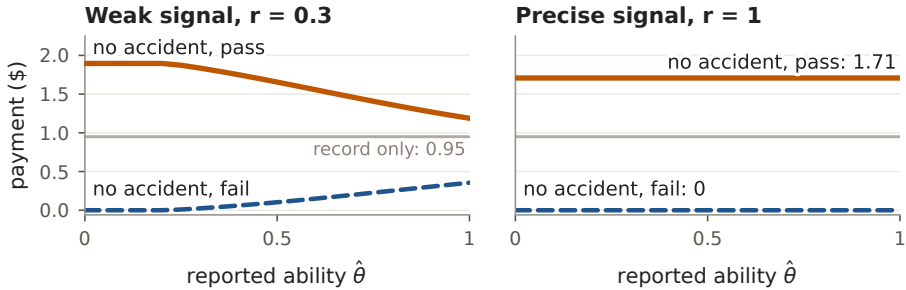
$$\varphi^*(r) \in \arg\max_{\varphi} V(r, \varphi).$$

Precision r	Best direction	Insurer saving	Cost reduction	Approx. $1.351\, r^2$
0.05	$-1.9°$	0.00338	0.10%	0.00338
0.30	$-1.5°$	0.133	3.9%	0.122
1.00	$0.9°$	0.788	23.1%	1.351

- **Direction:** within $2°$ of attention only for $r \leq 1$; it tilts toward ability as precision grows ($4.8°$ at $r = 2$).
- **Value:** the local approximation is exact to three digits at $r = 0.05$, 9% too low at $r = 0.3$, and 71% too high at $r = 1$.
- **Mechanism:** at each r, the insurer redesigns payments and recommended attention.

Monitoring changes the optimal contract

Payments by reported ability, along the best direction:



Weak signal: a menu. Reports up to 0.2 share a pass-only contract ($1.90); higher reports get flatter schedules.

Precise signal: one contract again, paying $1.71 after no accident *and* a pass, and zero otherwise.

How the new contracts change attention

In the benchmark, a common bonus induces $a^0(\theta)$. After a signal is added, each driver again chooses attention under the contract he selects.

More effective attention incentives

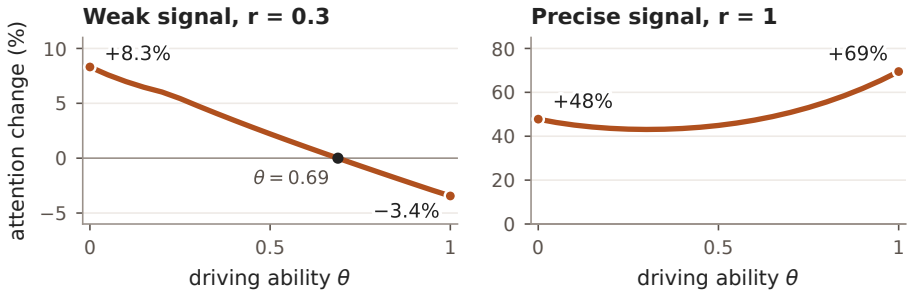
- Payment can depend more closely on attention.
- With less costly incentives, the insurer may want to induce a higher level of attention.

Different incentives across abilities

- A menu can offer different bonus schedules to different drivers.
- If ability were known, the optimal bonus would exceed the common bonus for less able drivers ($\theta < 0.55$) and fall below it for abler ones.

How monitoring reallocates attention

Attention relative to each driver's record-only level:



Weak signal: attention rises for $\theta < 0.69$ and falls above. Mean accident risk: 10.7% → 10.4%.

Precise signal: attention rises for every type, by 43% to 69%. Mean accident risk: 10.7% → 7.4%.

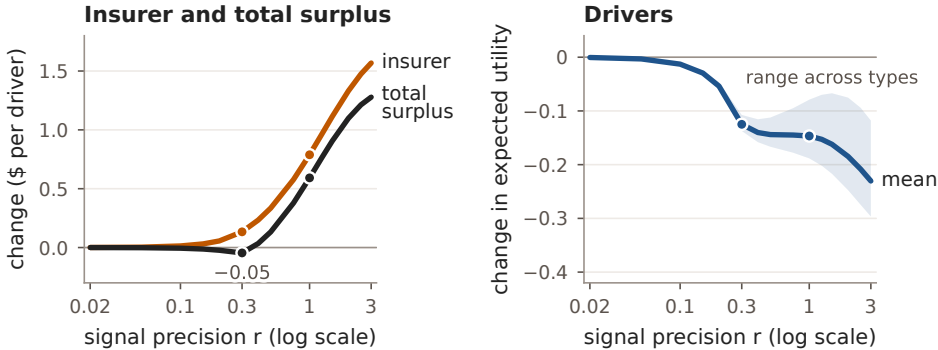
Weak monitoring reallocates attention; stronger monitoring raises it for everyone.

Who gains: rents, behavior, and payment risk

Monitoring changes three parts of the insurer's cost:

- **Driver rents.** Better targeted rewards can reduce extra driver utility. This benefits the insurer and can lower driver welfare.
- **Attention and accident losses.** More attention is costly but prevents accidents. Its money cost is the minimum certain compensation $h(c)$.
- **Payment risk.** Concentrating rewards on fewer outcomes can increase the cost of uncertain pay for risk-averse drivers.

Who gains in this calibration?



- **Weak signal ($r = 0.3$):** insurer $+0.13$, total surplus -0.05. Better attention saves 0.07, but extra payment risk costs 0.12.
- **Precise signal ($r = 1$):** insurer $+0.79$, total surplus $+0.59$. Better attention saves 0.74, which outweighs the extra payment risk cost of 0.15.

Takeaways

1. **What to measure depends on the incentive problem.**
   Here a weak signal should measure almost pure attention.

2. **Better monitoring changes who supplies attention.**
   A weak signal reallocates attention between different types of drivers.

3. **Welfare depends on behavior and payment risk.**
   The insurer gains and every driver loses. Total surplus falls with weak monitoring if payment risk dominates.

The general lesson

The value of additional information depends on the existing information and the economic incentives it changes.